OPERATOR GUIDE

Box3D Metal Quick Start

Reconstruct the pinned Box3D checkout, apply the Metal overlay, validate it, and enable GPU routing per world.

APPLE SILICON / METAL COMPUTE / BOX3D OVERLAY

Reconstruct and run

The public repository is an overlay: it contains no Box3D checkout. The bootstrap script clones the canonical upstream revision, verifies the patch, applies it, and builds a Release configuration.

Requirements

- Apple Silicon Mac and macOS Metal
- Xcode command-line tools
- Git, CMake, and Ninja

```
git clone https://github.com/Lulzx/box3d-metal.git
cd box3d-metal
./scripts/bootstrap.sh ../box3d-metal-worktree
```

Validation

```
../box3d-metal-worktree/build/metal-release/bin/test MetalTest
../box3d-metal-worktree/build/metal-release/bin/metal_demo
```

For a disposable end-to-end reconstruction, run `./scripts/verify-clean.sh`.

Enable per world

```
b3WorldId world = b3CreateWorld(&worldDef);
if (!b3World_EnableMetal(world, 32768)) {
/* CPU path remains usable. */
}
```

The body threshold is deliberately caller-selected. It is not a universal recommendation; measure your world with full-step timing.

Experimental residency stages

```
b3World_SetMetalFinalization(world, true);
b3World_SetMetalBroadPhase(world, true);
```

Finalization computes body and awake-shape results on Metal. Broad-phase mode currently moves raw dynamic-tree traversal only; CPU tree mutation, filtering, and contact creation remain.

Read route telemetry

```
b3MetalProfile p = b3World_GetMetalProfile(world);
printf("%s contacts=%llu pairs=%llu pairFallbacks=%llu\n",
p.deviceName, p.contactDispatchCount,
p.pairDispatchCount, p.pairFallbackCount);
```

Dispatch counters prove a Metal stage ran. Pair dispatch means raw tree traversal, not GPU tree ownership, filtering, narrow phase, CCD, sleeping, or events.

Next references

- docs/compatibility.md - exact supported and CPU fallback surface
- docs/performance.md - measured M4 Pro crossovers
- docs/troubleshooting.md - initialization, routing, and performance diagnosis